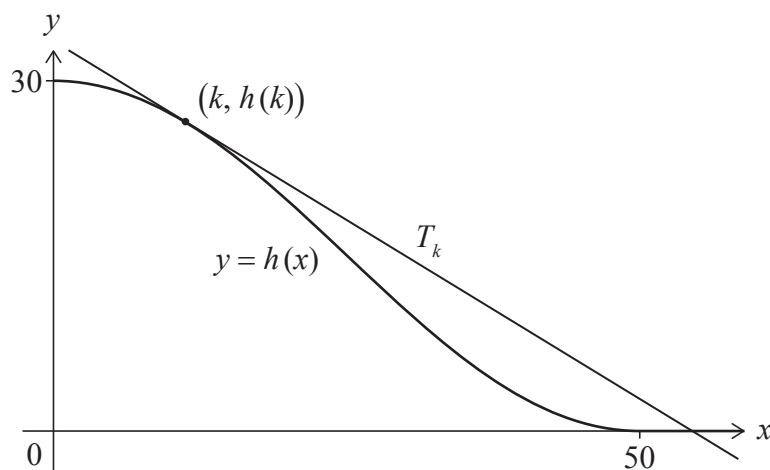


5. [Maximum mark: 6]

Consider the function $h(x) = 15\cos\left(\frac{\pi x}{50}\right) + 15$, where $0 \leq x \leq 50$.

The tangent, T_k , to the curve $y = h(x)$ at the point $(k, h(k))$ is shown on the following diagram.



- (a) Find the gradient of T_k in terms of k . [3]

Consider the case where the angle between T_k and the x -axis is $\frac{\pi}{8}$ radians.

- (b) Find the possible values of k . [3]

[illegible]